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low voltageI/OTouch screen control circuit

Overview

ET2046yes4Linear touchscreen controller, supports1.5V~5.5Vlow pressureI/Ointerface.ET2046With built-in2.5VThe built-in voltage source can be used for measurements in auxiliary input, battery monitoring, and temperature detection modes. When not in use, the built-in voltage source can be turned off to save power. The built-in voltage source can operate at a minimum of [insert value here].2.7VPower supply voltage, and can also be detected0V~6VBattery voltage.

becauseET2046It has low power consumption (at a power supply voltage of2.7VTime less than0.75mWHigh speed (maximum sampling rate can reach)125KHzFeatures such as its built-in chip driver make it a personal digital assistant with a resistive touchscreen.PDAs),BPIdeal for mobile phones and other portable devices.ET2046Can work in -40℃～85℃.

Features

pins andADS7846compatible

Operating voltage:2.5V~5.5V 1.5V

arrive5.25VnumberI/OBuilt-in

interface2.5Vvoltage source

Battery voltage can be measured directly.0V~6VOn-chip built-in

temperature measurement

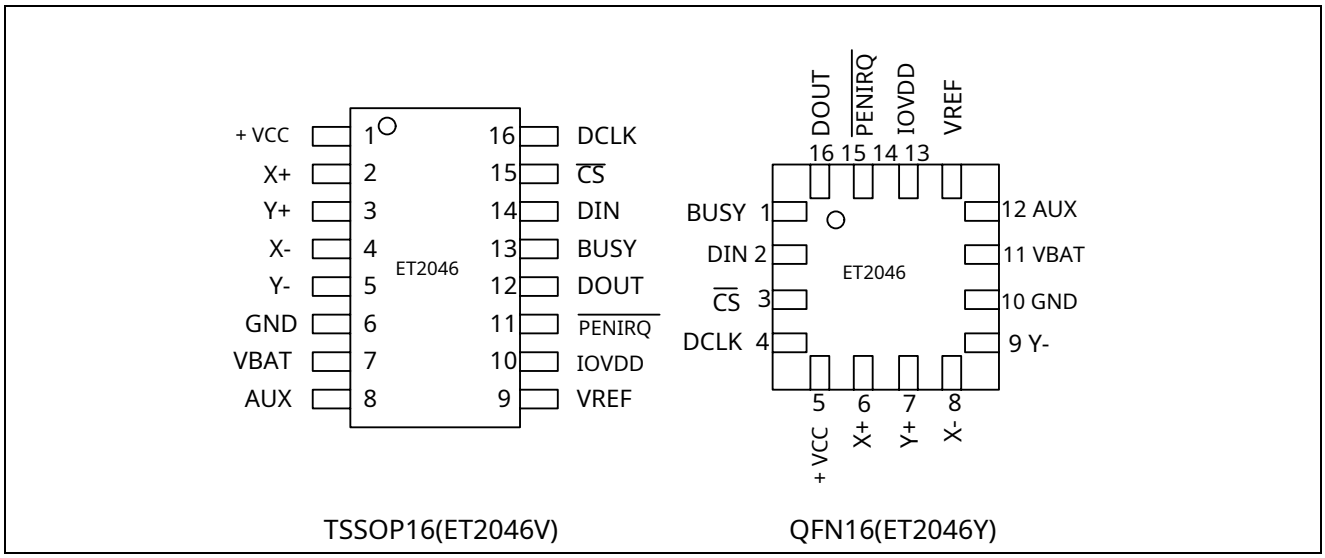
Touch pressure measurement

QSPI™andSPI™3Automatic power saving at

the wire interface

Packaging type:TSSOP16(ET2046V),QFN16(ET2046Y)

Pin arrangement diagram



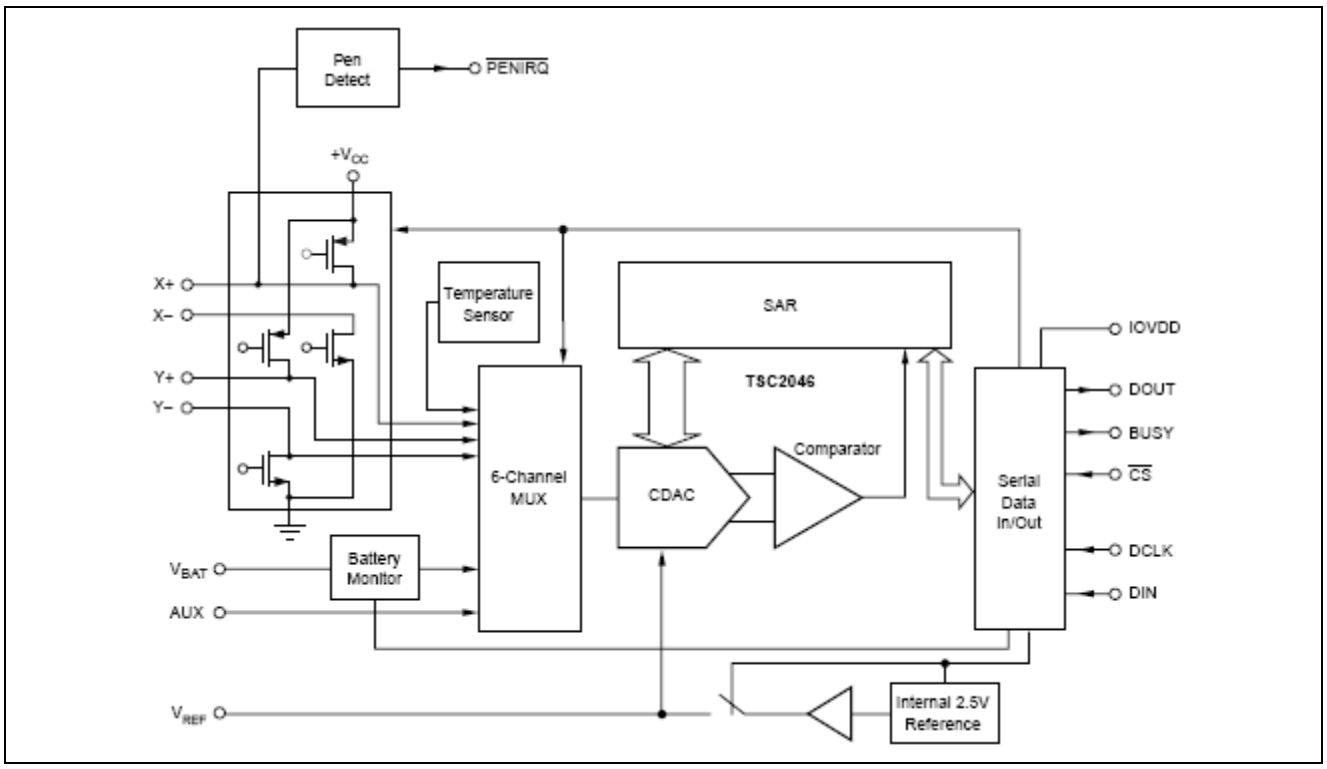
TSSOP16(ET2046V) QFN16(ET2046Y)

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Pin Description

Serial Number		Pin name	Function Description
TSSOP16	QFN16		
1	5	+ VCC	power supply.
2	6	X+	X+enter.
3	7	Y+	Y+enter.
4	8	X-	X-enter.
5	9	Y-	Y-enter.
6	10	GND	land.
7	11	VBAT	Battery monitoring input.
8	12	AUX	arriveADCAuxiliary input.
9	13	VREF	Voltage reference source input/output.
10	14	IOVDD	numberI/OPower input.
11	15	$\overline{\text{PENIRQ}}$	The stylus was interrupted.
12	16	DOUT	Serial data output. Data inDCLKThe falling edge is shifted out. When$\overline{CS}$When the value is high, this output is in a high-impedance state.
13	1	BUSY	BUSYOutput. When$\overline{CS}$When the value is high, this output is in a high-impedance state.
14	2	DIN	Serial data input. If...$\overline{CS}$The data is low.DCLKThe rising edge is latched into the register.
15	3	$\overline{\text{CS}}$	Chip select input. Controls conversion time and enables the serial input/output register.$\overline{CS}$For high =Power-Downmodel((ADC only).
16	4	DCLK	External clock input. This clock is used for...SARConversion process and synchronous serial dataI/O.

Functional block diagram



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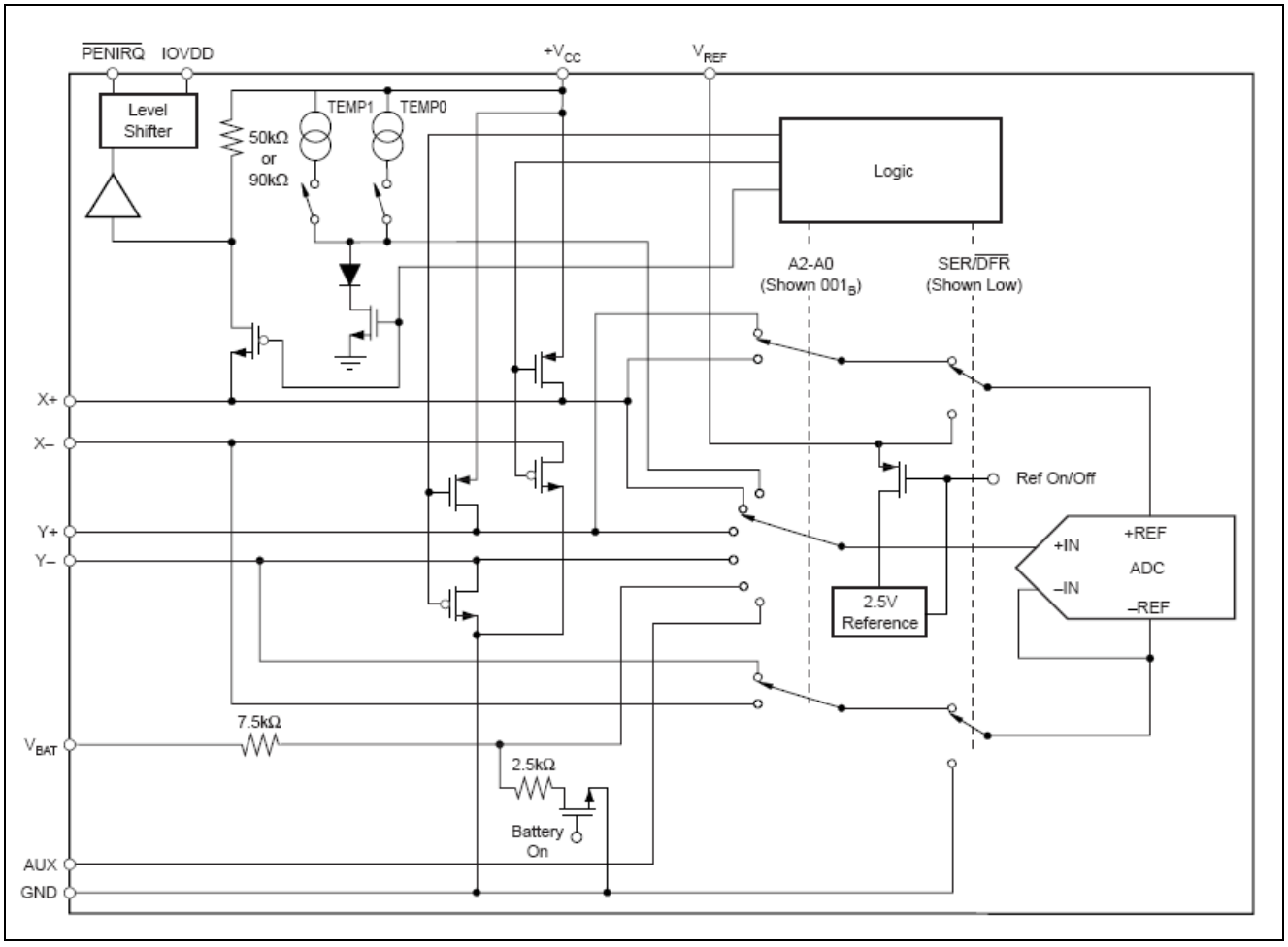
Function Description

ET2046It is a classic successive approximation register analog-to-digital converter (ADC).SAR ADCThis architecture is based on the principle of charge redistribution and inherently incorporates sample-and-hold functionality.

ET2046The basic working principle is shown in the figure below. This device has a built-in...2.5VThe voltage source, using an external clock, is available.2.7Varrive 5.25VThe internal power source can be powered by an external low-impedance power supply.1VTo +VCCIt is driven by a voltage source. The value of the source voltage directly determines the input range of the converter.

The analog input of the converter (X-,Y-andZCoordinates, auxiliary input, battery voltage, and chip temperature are provided via a multiplexer. A unique low on-resistance touchscreen driver switch allows an unselected...ADCThe input channel provides power to external devices, while another adjacent channel provides ground, such as a touchscreen. By maintaining the differential input and differential reference structure of the converter, the error caused by the on-resistance of the touchscreen driver switch (if this is a source of error under specific measurement conditions) can be reduced.

Analog Input



picture1Analog Input

The image above showsET2046Multiplexer on the deviceADCThe differential input and differential reference mode of the converter. Table1and table2ShowingA2,A1 ,A0andSER/DFRRelationships between control words andET2046Configuration. This control word is sent via serial port.DINProvided. When the converter enters hold mode, +INand-INThe input voltage difference is captured by an internal capacitor array. The analog input current is determined by the device's slew rate. During the sampling period, the source must sample the internal capacitor (typically [value missing]).25pFCharging. Once the capacitor is fully charged, there will be no more input current. The transfer rate from the analog source to the converter is a function of the conversion rate.

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A2	A1	A0	V_{BAT}	AUX_{IN}	TEMP	Y–	X+	Y+	Y-POSITION	X-POSITION	Z_1-POSITION	Z_2-POSITION	X-DRIVERS	Y-DRIVERS
0	0	0			+IN (TEMP0)								Off	Off
0	0	1					+IN		Measure				Off	On
0	1	0	+IN										Off	Off
0	1	1					+IN				Measure		X–, On	Y+, On
1	0	0				+IN						Measure	X–, On	Y+, On
1	0	1						+IN		Measure			On	Off
1	1	0		+IN									Off	Off
1	1	1			+IN (TEMP1)								Off	Off

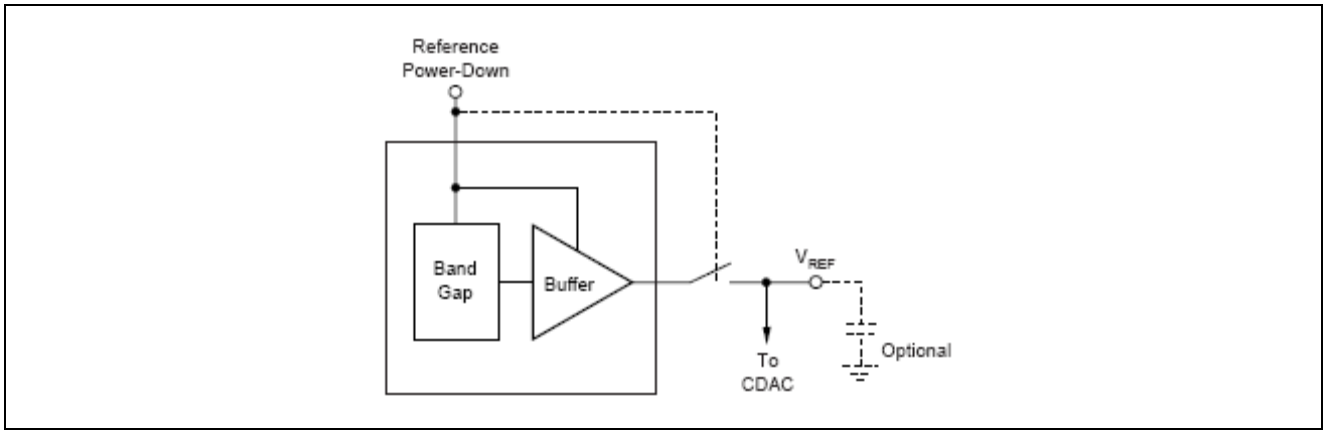
surface1Input configuration (DINSingle-ended reference source mode (SER/DFR(for high)

A2	A1	A0	+REF	–REF	Y–	X+	Y+	Y-POSITION	X-POSITION	Z_1-POSITION	Z_2-POSITION	DRIVERS ON
0	0	1	Y+	Y–		+IN		Measure				Y+, Y–
0	1	1	Y+	X–		+IN				Measure		Y+, X–
1	0	0	Y+	X–	+IN						Measure	Y+, X–
1	0	1	X+	X–			+IN		Measure			X+, X–

surface2Input configuration (DINDifferential reference source mode (SER/DFR(for low)

Internal Reference

ET2046Built-in one2.5VThe voltage reference source can be controlled by the control word.PD1Turn it on and off. This reference source is generally only used for battery monitoring, temperature measurement, and auxiliary input measurements in single-ended mode. Differential mode optimizes measurements of the touchscreen. For compatibility with...ADS7843For compatibility, the internal reference voltage source must be turned off. Therefore, a value must be written after power-on.PD1=0This ensures the source is closed.



picture2Simplified diagram of internal source

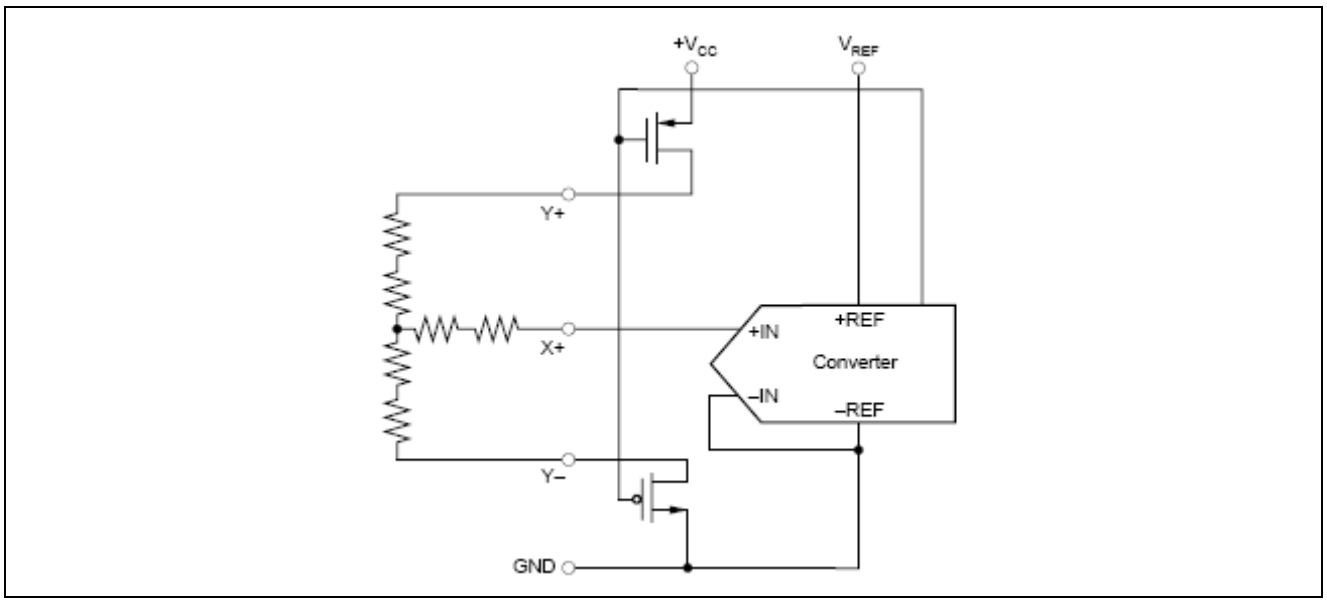
Reference Input

In +REFand-REFThe voltage difference between them determines the operating range of the analog input.ET2046exist1V~+V_{CC}It operates under a voltage source. Several key points related to the voltage source input and its wide voltage range require attention. As the source voltage decreases, the analog input value corresponding to each digital output code also decreases. This is related to the least significant bit (...).LSBRelated to)1LSBCorresponding to in12-bitSource voltage in mode divided by4096The value of . When the source voltage drops, due toLSBThe value also decreased, leading to thisADCThe inherent offset error and gain error will increase. For example, in 2.5VThe offset error of the converter under voltage source is2LSBs, and1VIts error can reach [percentage] under the voltage source.5LSBsHowever, in both cases, the absolute value of the device offset error is the same, which is 1.1.22mVAt lower reference voltages, the layout must be carefully designed, with sufficient filter capacitors added, and low-noise, low-voltage components used.rippleIf an external voltage reference source is used, a low-noise reference source must be used, and a low-noise input signal must be used.

enterV_{REF}The voltage at the port directly drives the digital-to-analog converter (DAC).CDACThe capacitor portion of the circuit has extremely low input current (typically < 1000 rpm). 13μAHere are a few key points to note regarding the reference source when shielding measurements are performed but the switch drive is on. Please refer to the figure for illustrative purposes.1This application diagram illustrates...ET2046Used for measuring resistive touchscreens. For measuring equipment in...YThe current value in the direction needs to be...X+Input connectedADC, OpenY+andY-Drive, RequantizationX+Voltage on (see figure)3(See the module diagram shown). For this measurement,X+Lead resistance does not affect the conversion (it affects settling time, but this resistance value is generally very small and its effect is negligible). However, becauseY+andY-The resistance between them is quite low.YThe on-resistance of the driver will have some impact. In summary, regardless of where the touchpoint is positioned on the touchscreen, the input cannot be...0VOr the full range, because some voltage is already lost in the internal switches. Additionally, the internal switching resistor is not linked to the touchscreen's resistance, thus adding an additional...

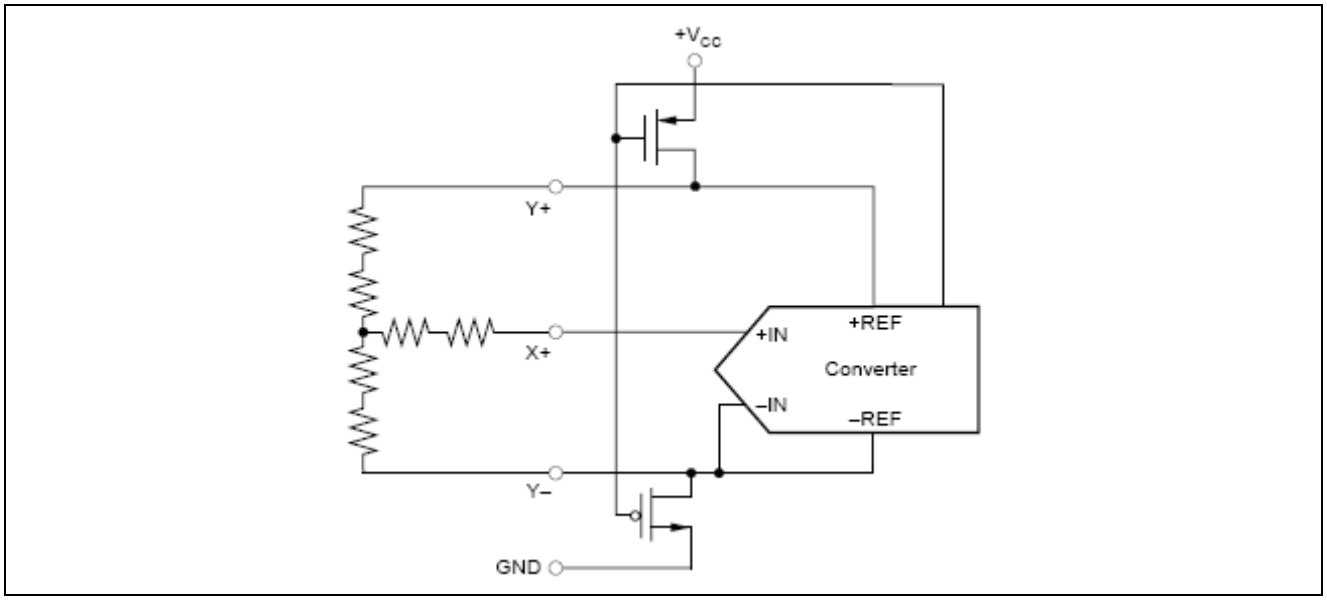
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External sources of error.



picture3Simplified block diagram of a single-ended reference source (SER/DFR = 1,YSwitch enable,X+(Simulated input) This situation can be illustrated by the diagram.5

The following is a remedy. In the placementSER/DFR = 0After, +REFand-REFDirectly connectedY+andX+This correspondingly puts the analog-to-digital converter into ratio conversion mode. The result of the conversion will be the percentage of the external resistance, regardless of changes in the ratio of the external resistance to the internal switch on-resistance. Note that power consumption should be considered when using ratio mode.



picture4Simplified block diagram of differential reference source (SER/DFR = 0,YSwitch enable,X+The last thing to note in differential mode (analog input) is that + must be used.V_{CC}InsteadV_{REF}As +REFThe source. When ratio mode is not required, a high-precision reference source and single-ended mode can be used for measurement. In certain situations, the converter can be started from a high-precision reference source. Most reference sources can be...ET2046It provides sufficient power, but may not be able to provide enough power for external loads such as resistive touchscreens.

The establishment of touch screen

In some cases, it may be necessary to connect capacitors across the touchscreen to eliminate noise generated during touchscreen operation (e.g., from the backlight circuitry or...). LCD(Noise generated on the panel) These capacitors provide a low-pass filter to reduce noise, but they introduce settling time when the screen is touched. The problem typically manifests as a gain error. Several methods exist to eliminate or mitigate this effect. The crux of the problem lies in the input and/or reference source...ADCThe sampling input and the provision of a digital output did not reach a final stable value. Additionally, the reference source may still experience issues during the conversion process.

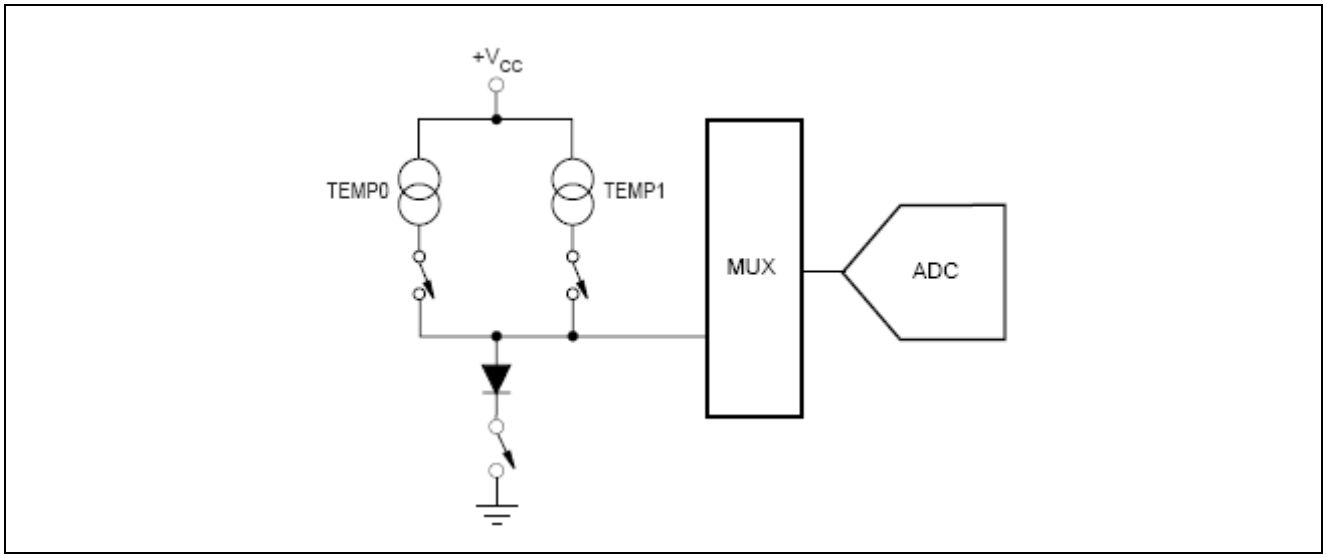
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It's changing. The first option is to stop or slow down within the required touchscreen setup time.ET2046ofDCLKSignal. This causes the input and reference source to be in the acknowledgment period (ET2046China3One clock cycle, see figure8This can achieve a stable value. It can be used in both single-ended and differential modes. The second option is to make...ET2046Only works in differential mode and configured when measuring the touchscreen.ET2046Keep it in working state (touchscreen driver enabled) and do not enterPower-Downstate(PD0=1According to the establishment time requirements andET20460The processor performs several conversions at a certain rate. Once the required number of conversions is reached, the processor commands...ET2046Enter after the final conversion.Power-Down Status. This process can be performed inXdirection,Ydirection andZThis is achieved through direction measurement. The third option is to make the device operate in...15In the mode of one clock cycle per conversion cycle, this allowsADCMaintain continuous operation by keeping the touchscreen driver always on until a stop command is received from the processor.

Temperature measurement

In some applications (such as when a battery is charging), it is necessary to measure absolute temperature.ET2046Temperature measurement technology originates from the characteristics of a semiconductor junction operating under a fixed current. The forward voltage of a diode junction (...)V_{BE}It has a strong correlation with temperature. In practical applications, it can be determined through known...25°CV_{BE}Value and monitorV_{BE}The absolute temperature can be obtained by calculating the offset value as the temperature changes. ET2046Two operating modes are provided. The first mode requires a voltage value at a known temperature as a standard, but only one measurement is needed to obtain the absolute temperature value. A diode (on) is used in this measurement process.20°C and have20μAWhen current flows through the diode, the typical value of this voltage is...600mVThe absolute value of the voltage across this diode will have...mVThe deviation of the level. However, the temperature coefficient of this voltage (TC) is very fixed, for

2.1mV/°C. In final product testing, to memorize this specific voltage used as a standard, the device will be stored in a room with a known room temperature. This method allows for measurement accuracy to reach [a certain level].0.3°C/LSB(exist12-Bit(in the mode).



picture5Schematic diagram of temperature testing mode

The second measurement method does not require testing a temperature standard, but it uses two temperature measurement processes to eliminate the influence of the lack of a temperature standard, thus achieving [the desired result].2The accuracy is °C. This mode requires a second conversion, during which the current flowing through the diode will be the same as the first conversion.91The pressure difference between the first and second transitions is given by the following formula (1)What it means:

$$\frac{KT}{q}\cdot\ln(N) \qquad (1)$$

in:

NThe ratio of currents =91

K=Boltzmann constant (1.38054·10$^{-twenty\ three}$(electron volt/Kelvin temperature)

q=Electronic charge (1.602189·10^{-19}C) T=Kelvin temperature

This method offers an improved way to measure temperature at the cost of reduced accuracy. The equation for solving the Kelvin temperature is:

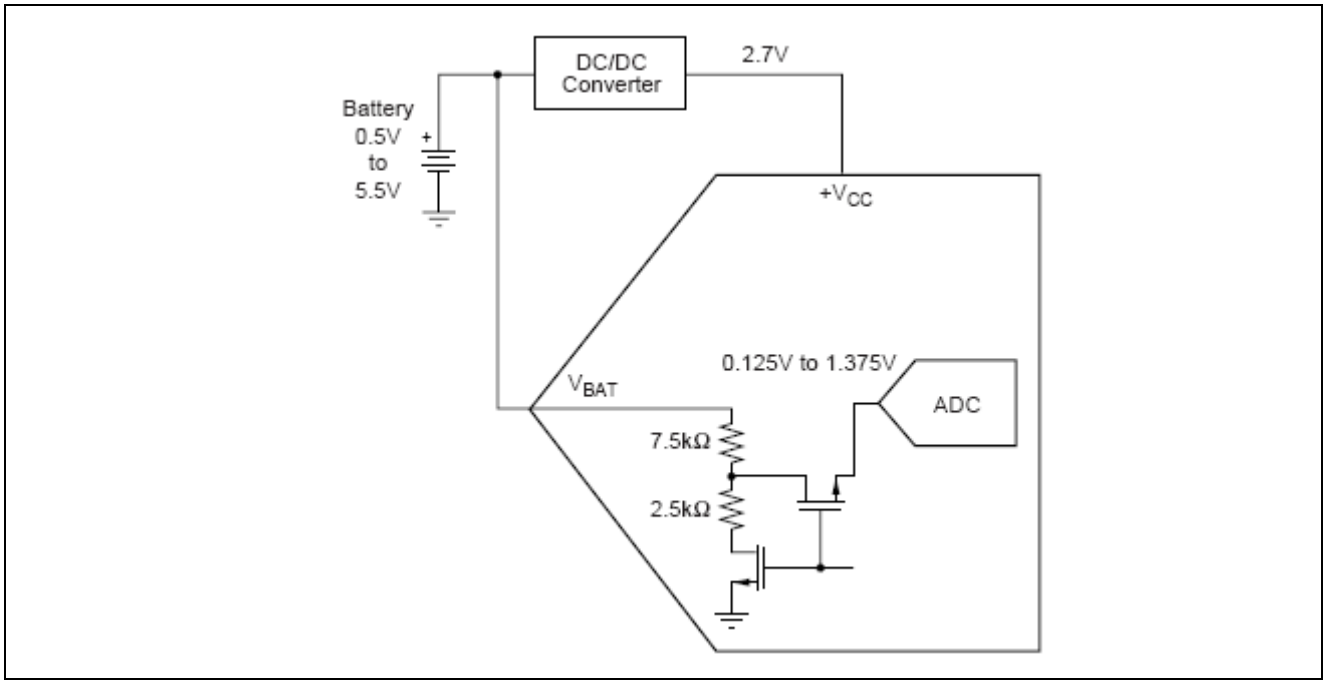
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$$\text{Kelvin temperature} = q \cdot \frac{\Delta V}{(k * \ln(N))} \qquad (2)$$

in:

△V=V(I91)-V(I1()(Units aremV) Kelvin temperature =2.573Kelvin/mV·△V ℃=2.573△V(mV)-273Kelvin **Battery voltage measurement**

ET2046With in voltage regulator (DC/DCThe converter's ability to monitor battery voltage on one side is shown in the figure.6The battery voltage can be from...0VChange to6VAt the same time, maintain supplyET2046The voltage is2.7Vor3.3VThe input voltage is divided by...4Therefore5.5VBattery voltage input ADC China is1.375VThis simplifies the multiplexer and control logic. To minimize power consumption, this voltage divider is only used in...A2=0,A1=1 andA0=0 It only works during the sampling period.



picture6Battery monitoring function block diagram

Pressure monitoring

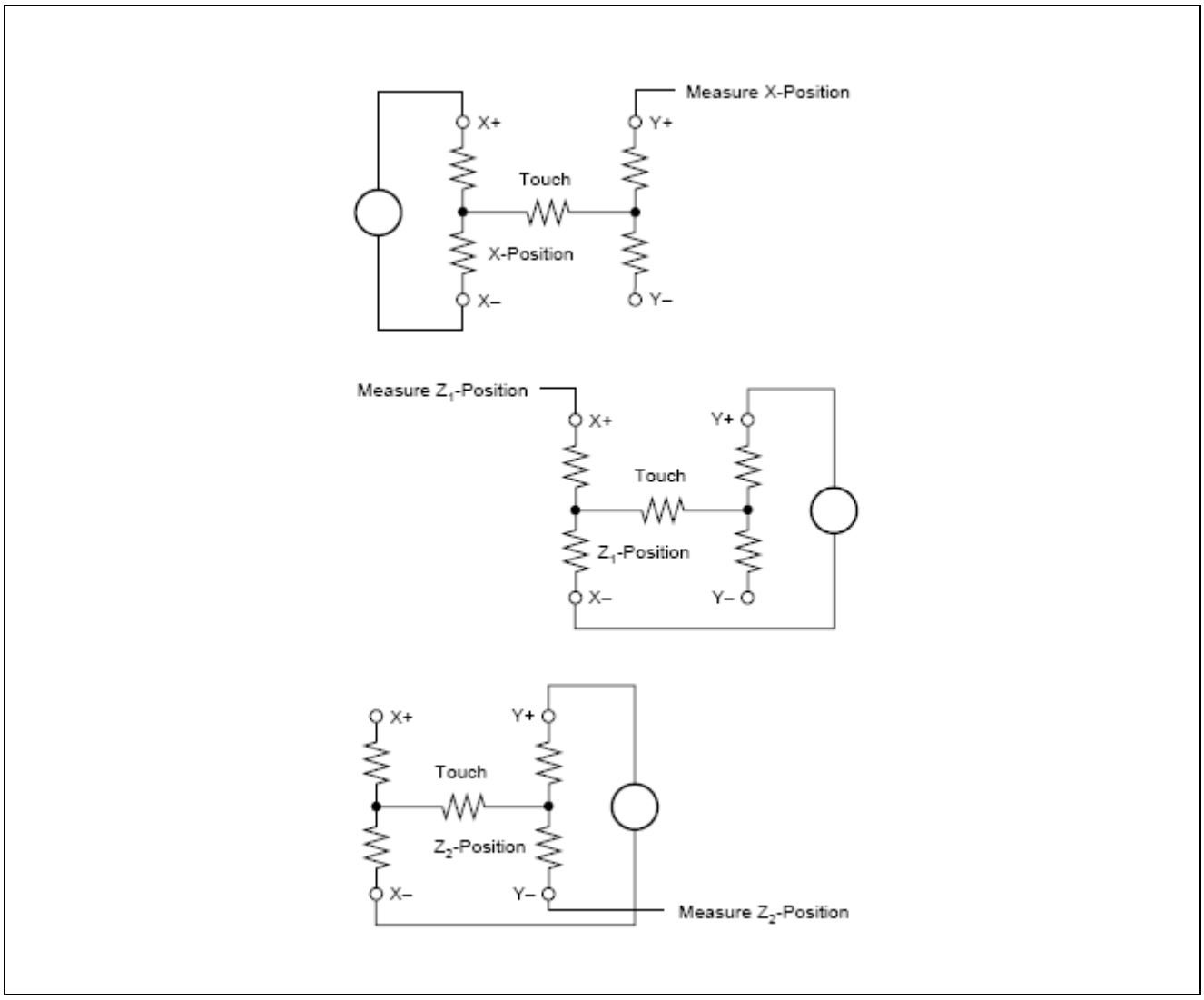
useET2046It can also monitor touch pressure. To distinguish between using a hand and a stylus to touch the screen, the pressure applied to the touch needs to be measured. Generally, this measurement does not require high precision, so using [a more precise measurement method] is recommended.8-BitPrecision mode (but the calculation process shown here is based on)12-Bit(Taking precision mode as an example). There are multiple methods to measure pressure.ET2046Two of these methods are supported. The first method requires known...XResistance of a plane, measurementXThe position was measured, and two additional plane positions on the touchscreen were also measured.Z1andZ2(as shown in the figure)7As shown. Using equation (3)The touch resistance can then be calculated.

$$R_{TOUCH} = R_{X-Plat} \quad e \cdot \frac{X - Position}{4096} \left(\frac{Z_2}{Z_1} - 1\right) \qquad (3)$$

The second method requires prior knowledge.Xplane andYResistance of a plane, measurementXdirection,YThe position of direction andZ1Using equations (4)Touch resistance can also be obtained:

$$R_{TOUCH} = R_{X-Plate} \cdot \frac{X - Position}{4096} \left(\frac{4096}{Z_1} - 1\right) - R_{Y-Plate}\left(1 - \frac{Y\text{-}Position}{4096}\right) \qquad (4)$$

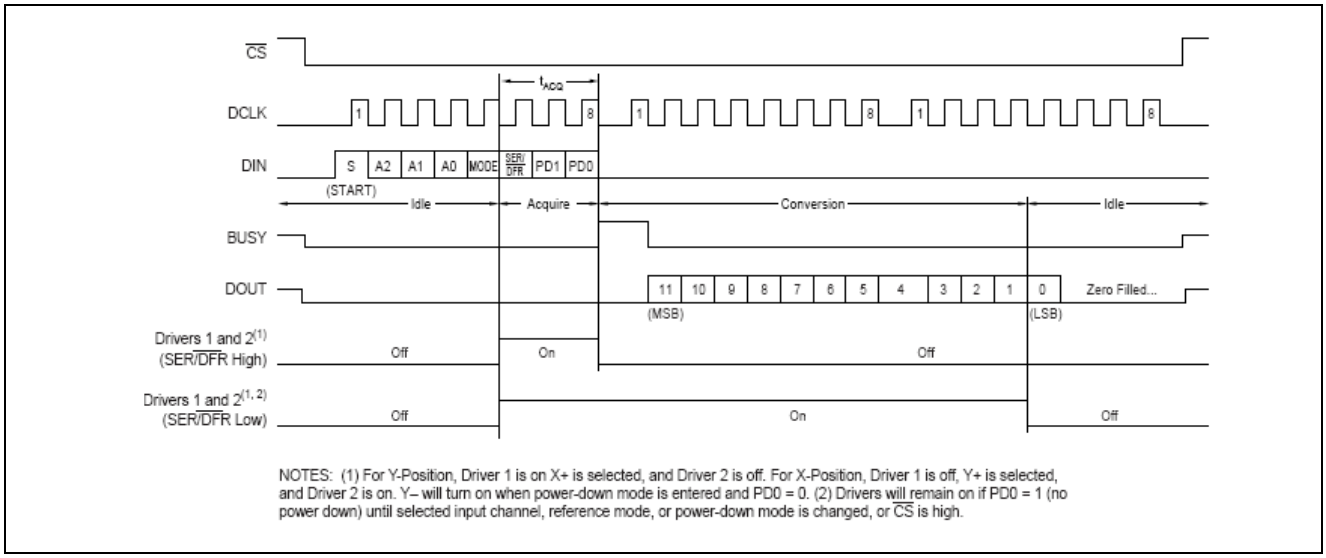
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picture7Block diagram of pressure measurement

Digital Interface

ET2046The typical working mode of a digital interface is shown in the figure.8.



NOTES: (1) For Y-Position, Driver 1 is on X+ is selected, and Driver 2 is off. For X-Position, Driver 1 is off, Y+ is selected, and Driver 2 is on. Y– will turn on when power-down mode is entered and PD0 = 0. (2) Drivers will remain on if PD0 = 1 (no power down) until selected input channel, reference mode, or power-down mode is changed, or $\overline{CS}$ is high.

picture8Conversion timing diagram, for each conversion cycletwenty fourOne clock,8Bit bus interface.

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This diagram assumes the digital signal originates from a microcontroller or digital signal processor with a serial port. Each communication between the processor and the converter, for example...SPI,SSIorMicrowire™Synchronous serial interfaces are all provided by8It consists of 1 clock cycle. A complete conversion can be made by 1 clock cycle.3The serial communication is complete.DCLKTotal inputtwenty fourOne clock cycle.

forward8One clock cycle is used to passDINThe port provides the control word. Once the converter receives enough information to configure the multiplexer and reference source input, the device enters an acknowledgment (sampling) mode, which will enable the touchscreen driver port if necessary. Then...3After one clock cycle, the control word transmission is complete, and the converter enters conversion mode. At this time, the input sample-and-hold mode enters hold mode, and the touchscreen driver port is turned off (in single-ended mode). The following...12The actual analog-to-digital conversion is completed in one cycle. If converted to ratio mode (SER/$\overline{DFR}$=0The driver port must remain open during the conversion process, and the second...13The final step is completed in one clock cycle.1 BitThe conversion requires additional steps.3The last byte is completed in one cycle.DOUT (The value will be low), these three cycles are ignored by the converter.

control word

Control word (added to)DINThe port providesET2046The following information is required: start conversion flag, address.ADCThe accuracy, configuration method andPower-DownThe mode selection is shown in the table.3.picture8,surface3and table4Detailed information is provided.

Start bit — the first bit (i.e.SThe start bit must be kept high to indicate the start of control word transmission. If the start bit is not detected...ET2046Will ignoreDINAll inputs on the feet.

Address bits—the next one3Bit(A2,A1,A0The input multiplexer is determined (see table).1,surface2and picture2The input channel for the touchscreen driver and the input for the reference source.

Mode bit — The mode bit is setADCThe precision. This bit is low, and subsequent conversions will use the same precision.12-BitThe mode is in progress; if this bit is high, the subsequent conversion will proceed in the following manner.8-BitThe mode is as follows.

SER/$\overline{DFR}$Bit-SER/$\overline{DFR}$This bit controls the mode of the reference source, specifying whether to use single-ended mode (high) or differential mode (low). Differential mode is also known as ratio mode and is used for measurement.Xdirection,YIt will outperform single-ended mode in both directional and touch pressure applications. The reference source voltage comes from the voltage of the switch driver, which is almost the same as the voltage applied to the touchscreen. In this case,ADCThe reference voltage is the voltage connected across the touchscreen, eliminating the need for a separate reference source voltage. In single-ended mode, the converter's reference voltage is always equal to...V_{REF} feet andGNDVoltage difference between pins (see table for details)1-2,picture1-4If measured in single-ended mode.Xdirection,YFor orientation and touch pressure, an external reference source is required. ET2046It must also be powered by an external reference source. Care must be taken when using single-ended mode.ADCThe input voltage must not exceed the voltage value of the internal reference source. This is especially true when the supply voltage is greater than...2.7Vhour.

Note: Differential measurement mode is only applicable to...Xdirection,YMeasurement of direction and touch pressure. Measurement of other values requires single-ended mode.

Bit7 (MSB)	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0(LSB)
S	A2	A1	A0	MODE	SER/$\overline{DFR}$	PD1	PD0

surface3The order of control bits in the control word

Bit	name	describe
7	S	Start bit. Control byte starts with first high bit on DIN.A new control byte can start every 15th clock cycle in 12-bit conversion mode or every 11th clock cycle in 8-bit conversion mode (see Figure 13).
6-4	A2-A0	Channel Select bits. Along with the SER/DFR bit, these bits control the setting of the multiplexer input, touch drivers witches, and reference inputs (see Tables I and II).
3	MODE	12-Bit/8-Bit Conversion Select bit. This bit controls the number of bits for the next conversion: 12-bits(low) or 8-bits (high).
2	SER/$\overline{DFR}$	Single-Ended/Differential Reference Select bit. Along with bits A2-A0, this bit controls the setting of the multiplexer input, touch driver switches, and reference inputs (see Tables I and II).
1-0	PD1-PD0	Power-Down Mode Select bits. Refer to Table V for details.

surface4Description of control bits in the control word

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PD0andPD1Bit-table5Describedpower-downThe configuration method of the mode and internal reference source. The internal reference source can be configured without affecting...ADCIt can be turned on and off under certain conditions. This allows the internal reference source extra time to establish its voltage value before the conversion takes place.ADC It works immediately without any additional setup time. To disable the internal reference source, it is necessary to...ET2046Write an additional instruction.

PD1	PD0	$\overline{\text{PENIRQ}}$	describe
0	0	Enabled	Power-Down Between Conversions. When each conversion is finished, the converter enters a low-power mode. At the start of the next conversion, the device instantly powers up to full power. There is no need for additional delays to ensure full operation, and the very first conversion is valid. The Y– switch is on when in power-down.
0	1	Disabled	Reference is off and ADC is on.
1	0	Enabled	Reference is on and ADC is off.
1	0	Disabled	Device is always powered. Reference is on and ADC is ON.

surface5 Power-DownSelection of mode and internal reference source

$\overline{\text{PENIRQ}}$Output

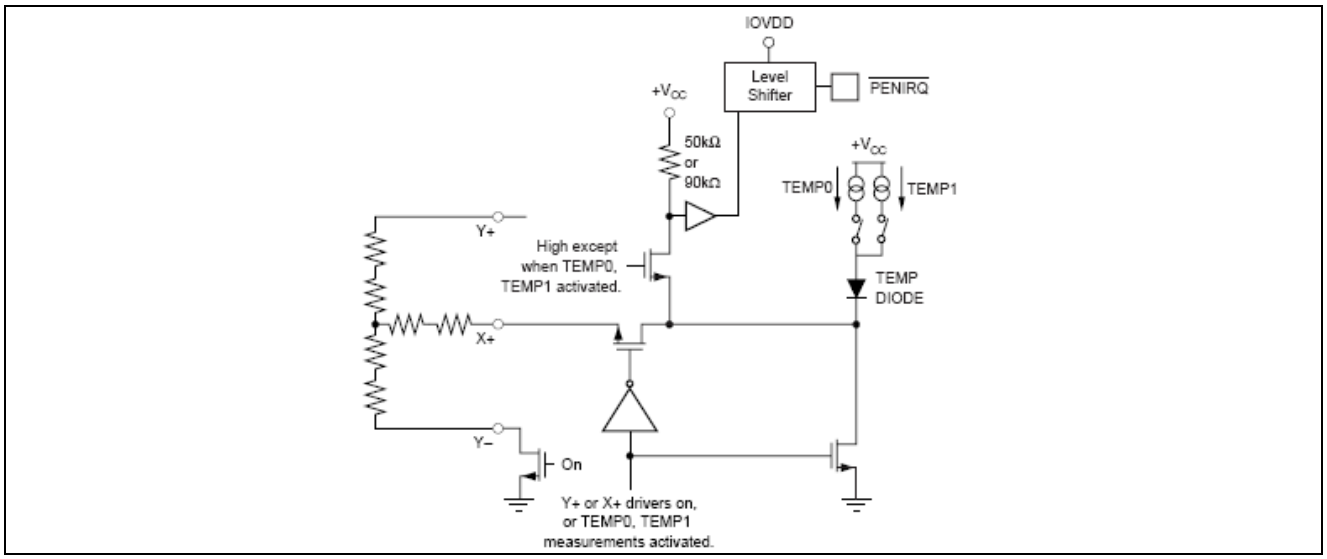
picture9This illustrates the output function of the pen interrupt. When inPower-DownIn mode andPD0=0hour,YThe drive will turn on the touchscreen.YDirectional grounding.$\overline{\text{PENIRQ}}$The output is connected to two transmission gates.X+Input. When the screen is touched.X+Input is made by dragging the touchscreen to the ground.

existET2046In the middle, the internal pull-up resistor is generally50kΩHowever, this value will change with temperature and process variations.36kΩand67kΩThe changes between them. To ensure...$\overline{\text{PENIRQ}}$Able to achieve0.35*(+Vcc)The logic low level is connected toX+andY-The total resistance between them must be less than21kΩ.

$\overline{\text{PENIRQ}}$A low level is achieved by draining current through the path from the touchscreen to ground, thereby generating an interrupt for the processor.X+, Y+andZThe measurement period of direction,X+Input and$\overline{\text{PENIRQ}}$The internal pull-up resistor is disconnected. This eliminates leakage current from the internal pull-up resistor through the touchscreen to ground (this current can cause measurement errors).

in addition,$\overline{\text{PENIRQ}}$Signal in measurementX+,Y+andZDirection is prohibited and kept low.$\overline{\text{PENIRQ}}$The signal is disabled and held high when measuring battery, auxiliary input, and temperature. If written...ET2046The last byte of the instructionPD0=1If the pen interrupt output function is disabled, it cannot detect whether the touchscreen has been touched. In this case, to re-enable this signal, the pen interrupt output function must be disabled.ET2046Write a tape PD0=0The control word. If written...ET2046The last control word containsPD0=0Then, the pen interrupt function will be enabled after this conversion is completed. The conversion ends when the converted data...Bit1SendET2046clockDCLKAfter the falling edge occurs.

We recommend sending control words to the processor.ET2046Time shielding$\overline{\text{PENIRQ}}$Interruption. This allows for the implementation of the scenarios discussed in this section.$\overline{\text{PENIRQ}}$ The output will not be triggered accidentally.

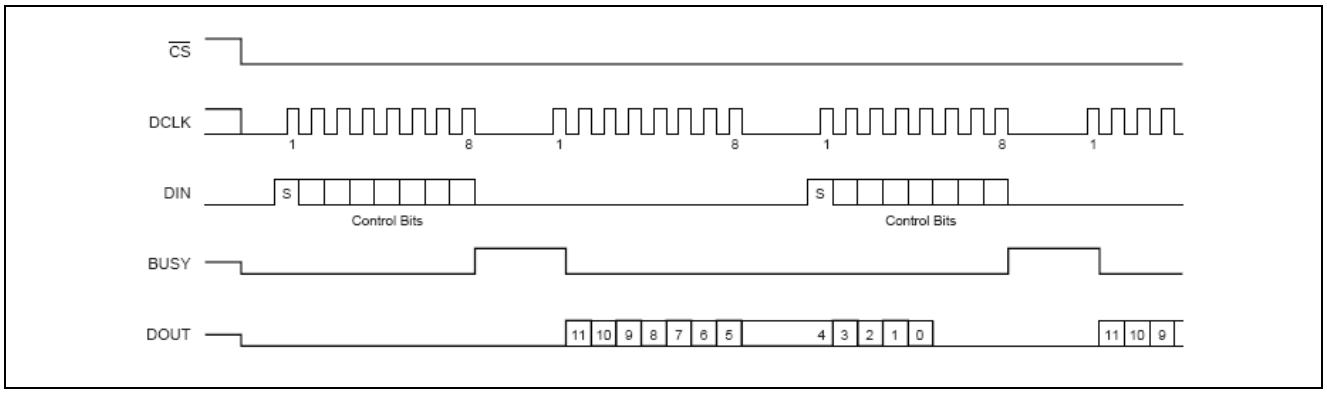


picture9 $\overline{\text{PENIRQ}}$Functional block diagram

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16Each clock cycle

As shown in the figure10As shown, conversionn+1The control word can then be converted.nInput, in order to achieve16One clock cycle per conversion period. This diagram also shows the format of the serial communication between the processor and the converter. From this diagram, we know that each conversion can begin after...1.6msComplete. Otherwise, the voltage across the sample-and-hold capacitor will drop significantly, affecting the conversion result.

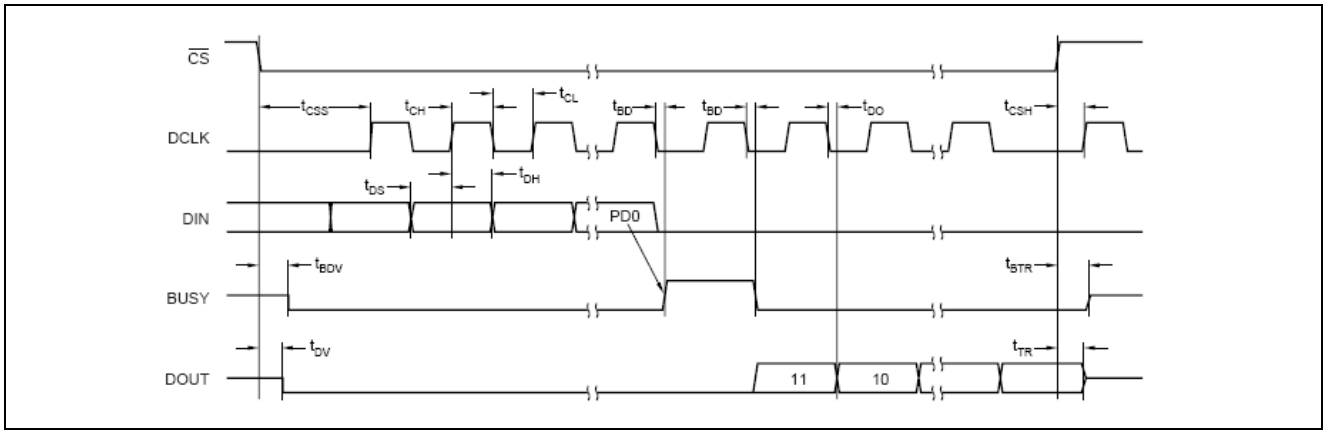


picture10Transform timing diagram,16Each clock cycle8Bit bus mode.

No need for specific serial portsDCLKDelay

Digital timing

picture8,picture11and table6ProvidedET2046Detailed timing information for the digital interface.



picture11Detailed timing diagram

symbol	describe	+ V_{CC}*2.7V,+V_{CC}*IOV_{DD}*1.5V,C_{LOAD}=50pF			unit
		Minimum value	Typical value	Maximum value	
t_{ACQ}	Acquisition Time	1.5			μs
t_{DS}	DIN Valid Prior to DCLK Rising	100			ns
t_{DH}	DIN Hold After DCLK High	50			ns
t_{DO}	DCLK Falling to DOUT Valid			200	ns
t_{DV}	$\overline{CS}$ Falling to DOUT Enabled			200	ns
t_{TR}	$\overline{CS}$ Rising to DOUT Disabled			200	ns
t_{CSS}	$\overline{CS}$ Falling to First DCLK Rising	100			ns
t_{CSH}	$\overline{CS}$ Rising to DCLK Ignored	10			ns
t_{CH}	DCLK High	200			ns
t_{CL}	DCLK Low	200			ns
t_{BD}	DCLK Falling to BUSY Rising/Falling			200	ns

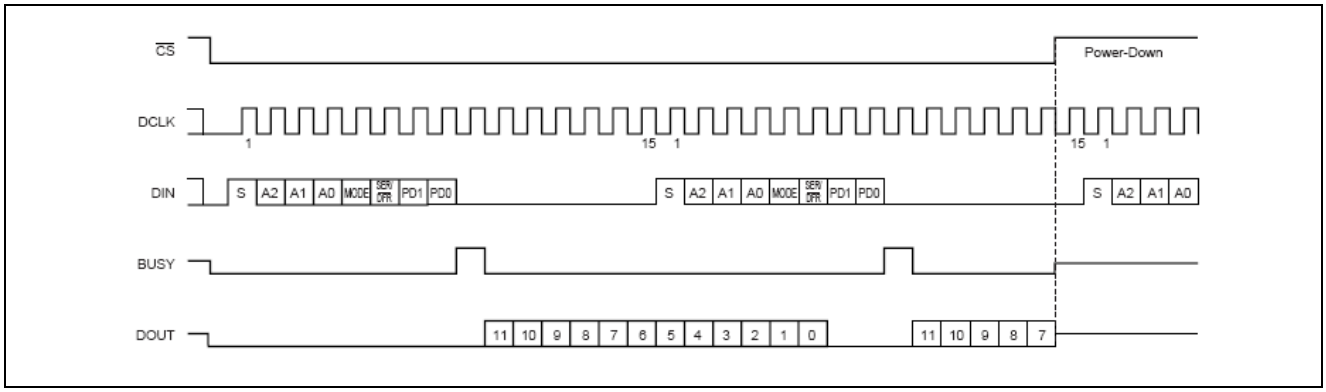
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t_{BDV}	$\overline{CS}$ Falling to BUSY Enabled			200	ns
t_{BTR}	$\overline{CS}$ Rising to BUSY Disabled			200	ns

surface6Timing values,Ta=-40℃ to +85℃

15Each clock cycle

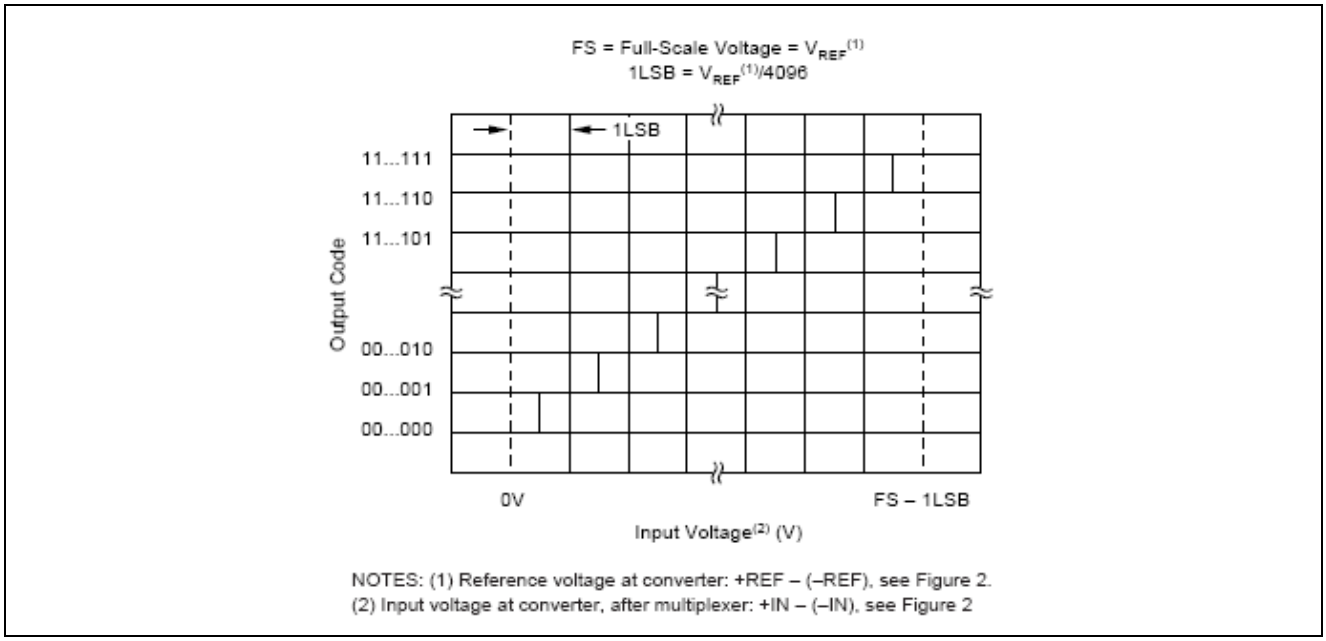
picture11ProvidedET2046The fastest clocking method. This method is not suitable for most microcontrollers and digital signal processors because they do not have the capability to provide clocking per signal transmission cycle.15The ability to operate on a single clock cycle. However, this approach is applicable to field-programmable gate arrays (FPGAs).FPGAs) or application-specific integrated circuit (ASICsNote that this method significantly increases the converter's maximum conversion rate.



picture12Maximum transmission rate15Each clock cycle

Data format

ET2046Output data directly in binary format, as shown in the figure.13As shown in the figure, this figure illustrates the ideal output encoding relative to a given input voltage, excluding the effects of offset, gain error, and noise.



picture13Ideal input voltage and output encoding

Switching modes

ET2046Provided8-bitThe conversion mode is designed for applications requiring high-speed conversion but not necessarily high precision in the digital results. When switching to...8-bitAfter the mode is set, each conversion can be done in advance.4This is completed in one clock cycle. This not only reduces the time required for each conversion...4Position (accelerate)25%At the same time, each conversion can occur at a faster speed. This is because...ET2046Internal setup time is no longer critical—it needs to be set to a higher level.8 The precision value of a bit. The clock rate can be increased to a maximum of [number] bits.50%Faster clocks and fewer clock cycles will increase the conversion rate by up to [percentage missing].1times.

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Limiting parameters

parameter	scope	unit
+ VccandIOVDDarriveGND	- 0.3~+6	V
Analog input to ground	- 0.3~+VCC+0.3	V
Number input to ground	- 0.3~IOVDD+0.3	V
Power consumption	250	mW
Maximum junction temperature	+150	℃
Operating temperature range	- 40~+85	℃
Storage temperature range	- 65~+150	℃
Wire temperature (welding,10s)	+300	℃

Electrical parameters(Unless otherwise specified, the test conditions are:)TA=-40℃~+85℃, +VCC=+2.7V,VREF=2.5VInternal voltage,fSAMPLE=125kHz, fCLK=16*fSAMPLE=2MHz,12-bitMode, Numeric Input =GNDorIOVDD,+VCCIt must be *IOVDD)

parameter	condition	Minimum value	Typical value	Maximum value	unit
Analog Input					
Full-range input	Positive input - Negative input	0		VREF	V
Absolute input range	Positive input	- 0.2		+ VCC+0.2	V
	negative input	- 0.2		+ 0.2	V
capacitance			25		pF
Leakage current			0.1		μA
System performance					
resolution			12		Bits
Missing code		11			Bits
Cumulative linearity error				±2	LSB(1)
Offset error				±6	LSB
Gain error	externalVREF			±4	LSB
noise	Including internalVREF		70		μVrms
Power supply rejection ratio			70		dB
Dynamic sampling					
Conversion time				12	CLK Cycles
Confirmation time		3			CLK Cycles
Channel throughput				125	kHz
Multiplexer setup time			500		ns
Aperture Delay			30		ns
Aperture jitter			100		ps
Channel isolation	VIN=2.5Vpp (50KHz)		100		dB
Switch driver					
On resistance					
Y+, X+			5		Ω
Y-, X-			6		Ω
Drive current(2)	Duration100ms			50	mA

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Reference source output					
Internal reference voltage		2.45	2.50	2.55	V
Internal reference source drift			15		ppm/℃
Static current			500		μA
Reference source input					
scope		1.0		+ V_{CC}	V
Input impedance	SER/$\overline{DFR}$ = 0, PD1 = 0		1		GΩ
	Internal source closed				
	Internal source enabled		250		Ω
Battery monitoring					
Input voltage range		0.5		6.0	V
Input impedance					
Sampling battery			10		kΩ
Battery monitoring off			1		GΩ
accuracy	V_{BAT}=0.5V~5.5V,external V_{REF}=2.5V	- 2		+2	%
	V_{BAT}=0.5V~5.5VInternal source	- 3		+3	%
Temperature measurement					
Temperature range		- 40		+85	℃
resolution	Differential mode(3)		1.6		℃
	TEMP0(4)		0.3		℃
accuracy	Differential mode(3)		±2		℃
	TEMP0(4)		±3		℃
Digital Input/Output					
Digital Logic Interface			CMOS		
capacitance	All digital control input pins		5	15	pF
V_{IH}	| I_{IH} | ≤+5μA	IOVDD*0.7		IOVDD+0.3	V
V_{IL}	| I_{IL} | ≤+5μA	- 0.3		0.3* IOVDD	V
V_{OH}	I_{OH}=-250μA	IOVDD*0.8			V
V_{OL}	I_{OL}=250μA			0.4	V
Data format			binary		
Power supply requirements					
+ V_{CC}(5)	Given work scope	2.7		3.6	V
	Scope of work	2.2		5.25	V
IOVDD(6)		1.5		+ V_{CC}	V
Static current(7)	Internal source closed		280	650	μA
	Internal source enabled		780		μA
	f_{SAMPLE}=12.5kHz		220		μA
	Power-Downmodel $\overline{CS}$=DCLK=DIN= IOV_{DD}			3	μA

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Power consumption	+ V_{CC}=+2.7V			1.8	mW
Temperature range					
Given working conditions		- 40		+85	℃

Note:

(1) LSBIndicates the least significant bit.V_{REF}=2.5Vhour,1indivualLSB=610μVDetermined during the design phase, but

(2) not tested. Exceeds...50mAExcessive current may cause device failure. MeasurementTEMP0andTEMP1The

(3) difference does not require a standard value. Temperature drift is -2.1mV/℃.

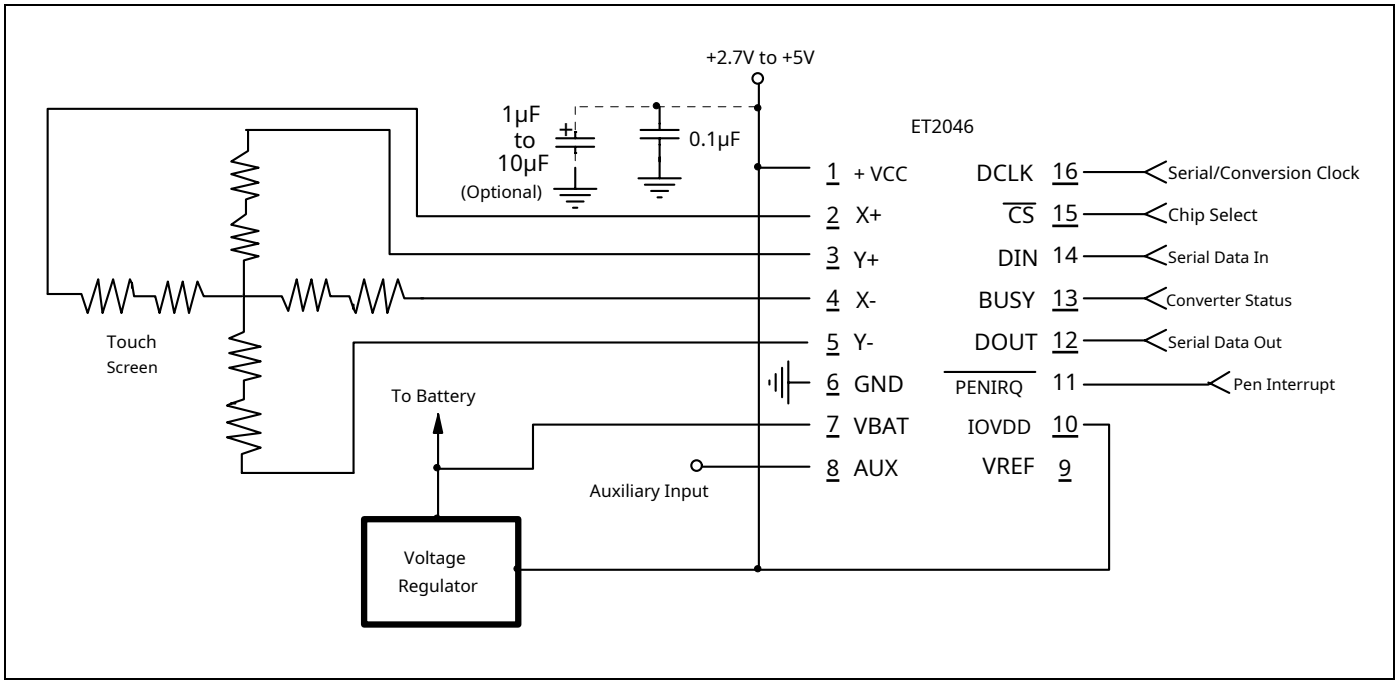
(4)

(5) ET2046Operating voltage can be as low as2.2V.

(6) IOV_{DD}It must be -(+)V_{CC}).

(7) Including +V_{CC}andIOV_{DD}The current, typically fromPD0=0hourAUXIt is obtained from the conversion of input.

Reference application circuit diagram

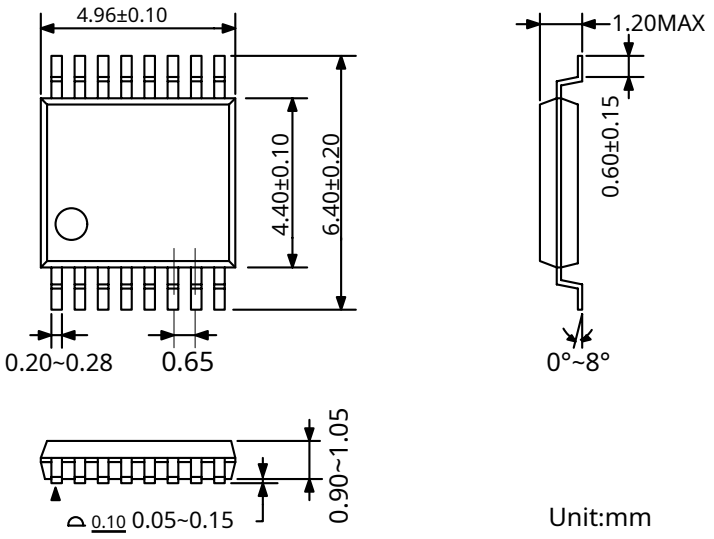


* : This circuit is for reference only.

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Package size

TSSOP16



QFN16

